AI101 Mini-Project

Predicting Bike Rental Demand Based on Weather and Time

Team 15 | School of AI and Data Engineering, IIT Ropar

Vantala Venu Prathyush
Execution Lead
2026AIB1090

Avishi Sood
Documentation Lead
2026EEB1330

Mirgane Aditya Tanaji
Video Lead
2026ESB1481

Pooja
PPT Lead
2026MCB1554

Repo: github.com/2026eeb1330/Bike-Rental-Demand-Prediction

Problem and data

What and why

• Predict how many bikes are rented each hour from weather (temp, humidity, wind, weather type) and time (hour, day, month, season).

• **Why:** operators can keep the right number of bikes at the right time.

Data details

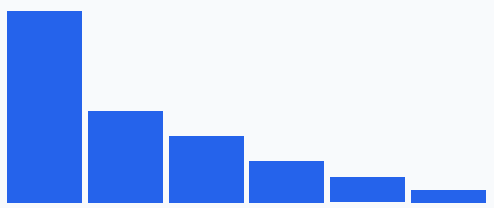
• **Data:** Kaggle Bike Sharing Demand, hourly records from Jan 2011 to Dec 2012.

• **Size:** 10,886 rows, 12 columns, no missing values (Kaggle test set: 6,493 rows).

• **Target:** count. casual and registered are dropped (they add up to count).

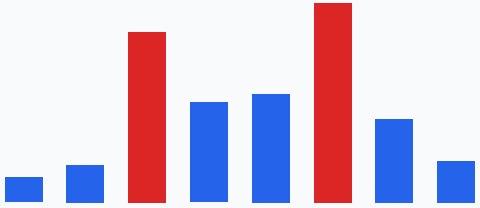
EDA Summary: demand peaks at 8 am and 5–6 pm, rises with temperature, and is right-skewed.

Distribution of Bike Rental Count



Rentals (Right-skewed)

Average Bike Rentals by Hour



Twin peaks at 8 AM & 5-6 PM

Bike Rentals vs Temperature



Higher temp = More rentals

Bike Rentals vs Humidity



High humidity reduces count

Approach

Data
train.csv (10,886 rows)

→

Preprocess
drop duplicates, date features, median fill

→

Split
80% train / 20% test

→

Model
Linear Regression

→

Output
predicted rentals + submission

13 Input Features

- **Weather:** season, weather, temp, atemp, humidity, windspeed
- **Calendar:** holiday, workingday
- **From datetime:** year, month, day, hour, weekday
- **Split:** random_state = 42 (8,708 train / 2,178 test)

Libraries & Constraints

- **pandas, numpy:** loading and cleaning data
- **matplotlib:** EDA and result plots
- **scikit-learn:** Linear Regression, train_test_split, MAE / RMSE / R^2
- **Predictions below 0 are set to 0 (rentals cannot be negative)**

Results: Linear Regression vs KNN

Linear Regression (baseline)

Train 8,708 | Validation 2,178

```
Shape of X: (10886, 54)
Shape of y: (10886,)
Shape of test features: (6493, 54)

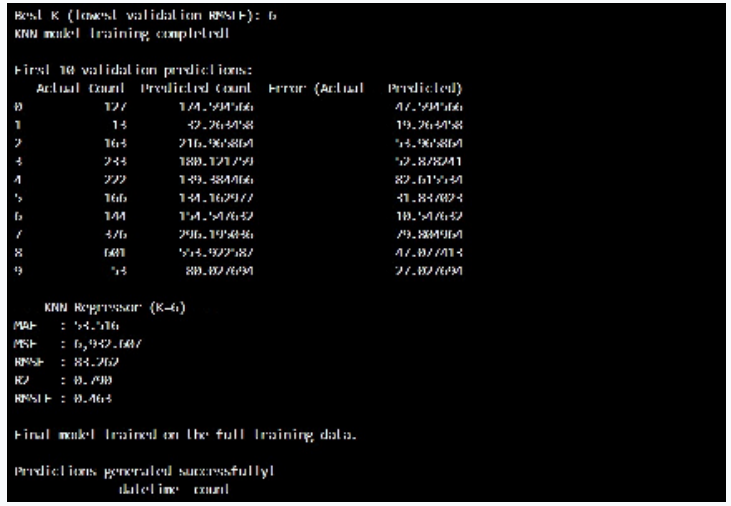
Training examples: 8708
Validation examples: 2178
Feature scaling completed.

--- Linear Regression (baseline) ---
MAE  : 68.710
RMSE : 97.214
R2   : 0.714
```

MAE	RMSE	R^2
68.710	97.214	0.714

KNN Regressor (K = 6)

Best K = 6 (lowest validation RMSLE)



MAE	RMSE	R^2	RMSLE
53.516	83.262	0.790	0.463

Key performance metrics

Here are our key performance metrics. Our Linear Regression baseline achieved an R-squared of **0.714** and an RMSE of **97.2**. Upgrading to KNN with K equals 6 boosted R-squared to **0.790**, dropped RMSE to **83.3**, and brought our validation RMSLE down to **0.463**.

Run from repo: `python model_training.py`

Learnings

What worked, what did not

Worked: full pipeline runs end to end; datetime features and EDA showed clear hour and temperature patterns

Did not: a straight line cannot fit rush-hour peaks ($R^2 \approx 0.40$)

Next: try tree-based models (e.g. Random Forest) and treat hour as a category

Who did what

- **Venu (Execution):** setup, requirements, code run
- **Avishi (Documentation):** GitHub repo, README
- **Aditya (Video):** 5-minute demo video
- **Pooja (PPT):** these 5 slides

All four wrote code and committed from their own accounts

AI-use disclosure

ChatGPT, Claude, Gemini: documentation and README improvement, help with code, debugging and slide drafting.

Model training, code verification, implementation and evaluation were done by the team; we ran the code and checked the numbers shown. No AI was used to generate the dataset or the results.